B4 mass-gap scope addendum – cone spectral gap, anchor identification, full digests

TECT verification-first repository · claim B4-MASS-GAP

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1. Purpose and scope

Operator verdict 2026-06-23 ACCEPTED the B4 evidence migration and left B4 at T5 (no tier action). This addendum records the three rigor items the verdict required before "mass gap" may be used as a physical statement: (i) the cone spectral gap Δ_{cone} is defined (§2); (ii) the anchor m^{*2} is recorded as a positive local curvature anchor *not identified* with Δ_{cone}, so no continuum/thermodynamic mass gap is claimed (§3); (iii) full SHA-256 digests replace the prefix matches (§4). No positive mass-gap claim is made here.

2. The cone spectral gap (tangent cone; operator review #2)

Let H_{Ψ_*} be the Hessian of F_{TECT} at the single-mode-cone critical point Ψ_*, let $T_{\Psi_*}\mathcal{C}$ be the *tangent cone* of the single-mode constraint set $\mathcal{C}$ at Ψ_*, and let $\mathcal{Z}$ be the antisymmetric zero modes. The admissible-variation space is the L^2 closure

$$\mathcal{H}_{\mathrm{cone}} := \overline{T_{\Psi_*}\mathcal{C} \cap \mathcal{Z}^{\perp}}^{L^2}, \qquad \Delta_{\mathrm{cone}} := \inf_{\substack{\delta\Psi \in \mathcal{H}_{\mathrm{cone}} \\ \|\delta\Psi\|_2 = 1}} \langle \delta\Psi,\, H_{\Psi_*}\, \delta\Psi \rangle. \tag{1}$$

The tangent *cone* (not a tangent space) is the rigorous object: the single-mode constraint set need not be a smooth manifold at Ψ_*. A genuine local mass gap requires $\Delta_{\mathrm{cone}} > 0$; hence $\Delta_{\mathrm{cone}} \le 0$ **is a direct falsifier of the gap**, distinct from the uniqueness falsifier.

3. The anchor is a curvature anchor, not the gap

The migrated continuation reports $m^{*2} = +4.247 \times 10^{-2}$ at $\mu^2 = +5 \times 10^{-3}$ on the metastable subset-4-cosine branch. This is a *positive local curvature anchor* within the single-mode cone. The record establishes **no** equality $m^{*2} = \Delta_{\mathrm{cone}}$ and **no** rigorous bound $m^{*2} \le \Delta_{\mathrm{cone}}$ (or $\ge$): the antisymmetric zero-mode-excluded lowest eigenvalue of H_{Ψ_*} is not computed in the migrated material. Consequently B4 is, in its current scope, a positive local-curvature statement, **not** a continuum or thermodynamic mass-gap proof. Establishing the latter requires (a) computing the symmetry-projected Δ_{cone} on the canonical production Hessian, (b) the identification or rigorous inequality relating m^{*2} and Δ_{cone}, (c) the continuum ($N \to \infty$ / $a \to 0$) limit, and (d) the production-Hessian reproduction bundle (PACKAGE-PENDING).

4. Full SHA-256 digests (provenance, prefix match insufficient)

```
6aeffab92c313edc9223748d4562581d33956b8cb856eea29f5c527ce427b45e
  Math01/TECT-Math01-v2-BCC-uniqueness-rigorous.tex.txt
26255775038bdc347f6eeeccbc83383a1d13e6bfffe26472afc8453c4d7a4601
```

```
  Math56/TECT-Math56-AddB-ClassII-guarded-quotient-analytical.tex.txt
a5546466a8bc4cdf8255746e5196ce2cca0070f1b30fb387b4736f193ba518d6
  Math56/TECT-Math56-Addendum.tex.txt
4c99f5f9c7cfb9f3c0ed8dcfbd4ded62429ea44ca70414fc70f8077417f5a3b4
  Math56/TECT-Math56-HessJump-audit.tex.txt
9c76de35837b370aec945921f120512d145780b9e16780dc229266d1b0028b64
  Math82/TECT-Math82-Addendum-G-Phase-Z-7point-bifurcation-curve.tex.txt
2025a6655421af147d8312c410d8af5e3473b25c9d764a64723f623f42a5b911
  Math82/TECT-Math82-Addendum-G2-PCG-and-stall-mechanism-audit.tex.txt
fd038f4a485e14699113f37e552bc5615200e96dbd5a522cd2bb67b018d71776
  Math82/TECT-Math82-Addendum-G3-vacuum-floor-guard-implementation.tex.txt
```

These full digests are also recorded in `archive/MIGRATION-LEDGER.md` (batch 3). They authenticate byte-identity of the archive copies against the frozen legacy originals.

5. Operator verdict (2026-06-23)

Item	Operator decision
B4 evidence migration	ACCEPTED
B4-MASS-GAP	$T5 \to T1$ OPEN/UNESTABLISHED
B4-CONE-CURVATURE-ANCHOR	new T5 (curvature anchor)
Numerical provenance	PACKAGE-PENDING
Full physical mass gap	NOT established ($m^{*2} > 0 \not\Rightarrow \Delta_{\text{cone}} > 0$)
Continuum / production-Hessian gap	still open

6. Devil's-advocate (three objections)

(α) *"A positive m^{*2} already proves a gap."* DISMISSED: m^{*2} is the curvature in the single-mode direction at a metastable anchor; the gap is the symmetry-projected infimum Δ_{cone} over the whole cone, which is not computed. Positivity of one curvature does not bound the infimum.

(β) *"Prefix SHA-256 was good enough."* UPHELD: a 8-hex prefix is 2^{32} collision space; the full 256-bit digests (§4) are now recorded, closing the authentication gap.

(γ) *"The metastable, $\Delta F > 0$ anchor is the wrong object for a mass gap."* VALID with mitigation: the record states the anchor is metastable and unfavourable; the addendum makes explicit that no mass-gap claim is attached to it. The favourable point (#3) is also migrated.

Quantitative sanity check (mandatory): the seven full digests are 64 hex characters (256 bit) each and reproduce by `sha256sum` on the archive copies; the prefix-to-full extension is verified to agree on the leading 8 hex ($6aeffab9\ldots$, $26255775\ldots$, etc.).

7. Result footer

```
Result ID:              B4-MASS-GAP (reclassification addendum)
Precise statement:      The mass-gap claim is not established:
                        m^{*2}>0 does not imply Delta_cone>0.
Scope:                  B4-MASS-GAP is OPEN/UNESTABLISHED.
Dependencies:           B4 migration record v1.0; A1-KERNEL-CONV.
Evidence grade:         ANALYTIC (scope) + provenance (full digests).
Reproduction command:   sha256sum archive/legacy/notes/Math0{1}/* \
                        archive/legacy/notes/Math56/* archive/legacy/notes/Math82/*
Expected output:        the seven digests of Section 4.
Falsification gate:     (gap) Delta_cone<=0; (uniqueness) a second cone
```

```
                         minimiser. The two are distinct.
Tier before / after:     B4-MASS-GAP T5 / T1 OPEN.
Successor:               B4-CONE-CURVATURE-ANCHOR, T5:
                         metastable single-mode curvature anchor only.
No-overclaim statement:  No cone spectral gap, no continuum mass gap,
                         and no thermodynamic mass gap are established.
Next required action:    Production-Hessian Delta_cone computation +
                         m*^2--Delta_cone relation + continuum limit +
                         reproduction bundle, to upgrade beyond the curvature
                         anchor.
```